

SOLUTION ARCHITECTURE DOCUMENT

PFZ-AI

Document Control

Version History

Version	Date	Author	Key changes
Ver1	16-02-20216	Renny	First Draft for Review

Document References

#	Title	Link

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1. Introduction

1.1 Document Purpose

The purpose of this document is to describe the IT solution design for the PFZ-AI

The purpose of the solution design is to:

- Communicate the end-to-end IT solution to all stakeholders
- Provide traceability of the solution back to business requirements and reference architecture
- Provide all views of the solution required for design, build, testing and implementation
- Define impacts of the solution for estimation, planning and delivery purposes

The structure of the document develops the solution from the drivers of why the solution is needed, through the conceptual non-technical description to the solution design and delivery implications.

- Drivers and constraints of the solution – To define the purpose and goals of the project along with the constraints of the solution
 - Project overview
 - Reference architectures
 - Business context
 - Requirements
- Conceptual solution - Defining the conceptual architecture by defining the current and target state in a more abstract view
- Logical solution – To specify the solution using the various architecture views
- Solution delivery - Specify the delivery approach and assessment of the solution
 - Delivery considerations
 - Architecture assessment
 - Impacts assessment

1.2 Document Scope

The scope of this document is limited functional and non-functional requirements and processes related to the *[project]*.

1.3 Document Audience

The audience for this document includes:

- **Project Leadership Team:** Renny, Jayaram, Joshua, Herral , Shyam
- **Product Team:** Herral
- **Sales Team:** Shyam , Biplab
- **Partnership – Jayaram**
- **UI/UX – Bobbin**
- **AI/GenAI – Anish**
- **Backend – Joshua , Ezhil**

1.4 Definitions

The following acronyms and definitions will be used in the *[project]* and within this document.

Term/Acronym	Definition
PFZ	Potential Fish Zone
GenAI	Generative AI

2. Solution Overview

The present invention generally relates to the field of marine technology leveraging Artificial Intelligence (AI), Generative AI (GenAI), and Agentic AI to forecast Potential Fish Zones (PFZ) across the sea coast. The system utilizes historical weather data and PFZ data to predict future PFZ and incorporates automated feedback for continuous learning. Fishermen setting out from harbours require prior knowledge of Potential Fish Zones (PFZ) to plan their fishing trips efficiently. Advance knowledge of PFZ helps in optimizing routes, reducing expenditure on fuel, ice, and other resources, and improving overall profitability while minimizing operational costs. Existing solutions rely primarily on satellite technologies and real-time data, which do not provide sufficient advance notice (at least one day ahead) for effective planning. Additionally, such solutions lack customization based on boat-specific parameters such as engine power, storage capacity, fishing net type, and operational depth. Current systems also do not leverage historical data effectively to improve prediction accuracy and increase the likelihood of better catch outcomes. Furthermore, existing PFZ advisories are typically delivered via websites, making them inaccessible to fishermen at sea where connectivity is limited or unavailable.

There exists a need for a solution that can predict Potential Fish Zones (PFZ) a few days in advance using historical data to enable better planning and optimal utilization of resources. There is also a need for customized predictions based on boat characteristics and for ensuring availability of such predictions at sea through satellite connectivity. Further, there exists a need for a system incorporating continuous learning by feeding prediction outcomes back into the system to enhance accuracy over time.

2.1 Solution Objectives

A primary objective of the present invention is to provide a system for predicting Potential Fish Zones (PFZ) in advance using historical weather and PFZ data. And enable fishermen to plan fishing trips efficiently by identifying PFZ locations prior to departure.

Another objective of the present invention is to develop a customized PFZ prediction mechanism based on boat-specific parameters such as power, storage capacity, fishing nets, and operational depth. and ensure availability of PFZ predictions to fishermen at sea through satellite-based connectivity. PFZ-AI also aims to incorporate Artificial Intelligence, Generative AI, and Agentic AI for PFZ forecasting.

Another objective of the present invention is to integrate multiple data sources including satellite data, historical weather data, and third-party PFZ providers. This will help to provide coast-specific PFZ predictions considering localized climatic and environmental conditions and optimize utilization of resources such as fuel, ice, and operational costs for fishermen.

2.2 Solution Scope

The present invention provides a PFZ prediction system based on historical weather and climatic data using GenAI and Agentic AI technologies. It generates customized PFZ forecasts considering parameters such as boat characteristics, depth, climatic conditions, and coastal regions.

The system ensures delivery of predictions to fishermen in deep sea through satellite communication. It incorporates a continuous learning mechanism that compares predicted and actual outcomes, enabling improved accuracy over time. The system also identifies gaps between prediction and reality and uses these insights to refine future forecasts.

The arrangement of the present invention has other features and advantages which will be apparent from or are set forth in more detail in the accompanying drawings, which are incorporated herein, and the following brief description, which together serve to explain certain principles of the present invention.

2.2.1 In Scope

The functional scope for the project includes:

- *Data Sourcing from various websites – INCOIS*
- *Data Transformation*
- *Analytics aided PFZ Forecasting*

- *Mobile UI Development for PFZ*
- *Admin Screens Development*
- *Cloud Configuration and hosting*
- *Coast specific Agentic AI Development*

2.2.2 Out of Scope

The following are deemed out of scope

- *Integration with Boat hardware*
- *Species specific forecast*
- *Non-Indian Coast forecast*
- *Network Connectivity*

2.3 Assumptions

The following assumptions will be validated as part of solution design and implementation:

- The forecasting assumes that there is a correlation between weather parameters and PFZ
- The forecast also assumes that there are cyclic patterns for fish availability across the coasts

2.4 Constraints

The following constraints apply

- PFZ-AI requires a Satellite Phone and Network Connectivity to be able to access the predictions

2.5 Dependencies

The solution is dependent on the following items:

- Network Connectivity
- Real clients who can validate the predictions
- INCOIS PFZ predictions for Realtime predictions

2.6 Key Architecture Decisions

#	Title	Decision
AD1	AI-First, API-First	Core capabilities exposed via APIs for interoperability and future-proofing.
AD2	Extensible by Default	Modular and pluggable architecture for easy tailoring and integrations.
AD3	User Empowerment	Visual and guided tools for both novice and advanced users.

AD4	Operational Excellence	Automated monitoring and self-healing for reliable, scalable operation across te coasts

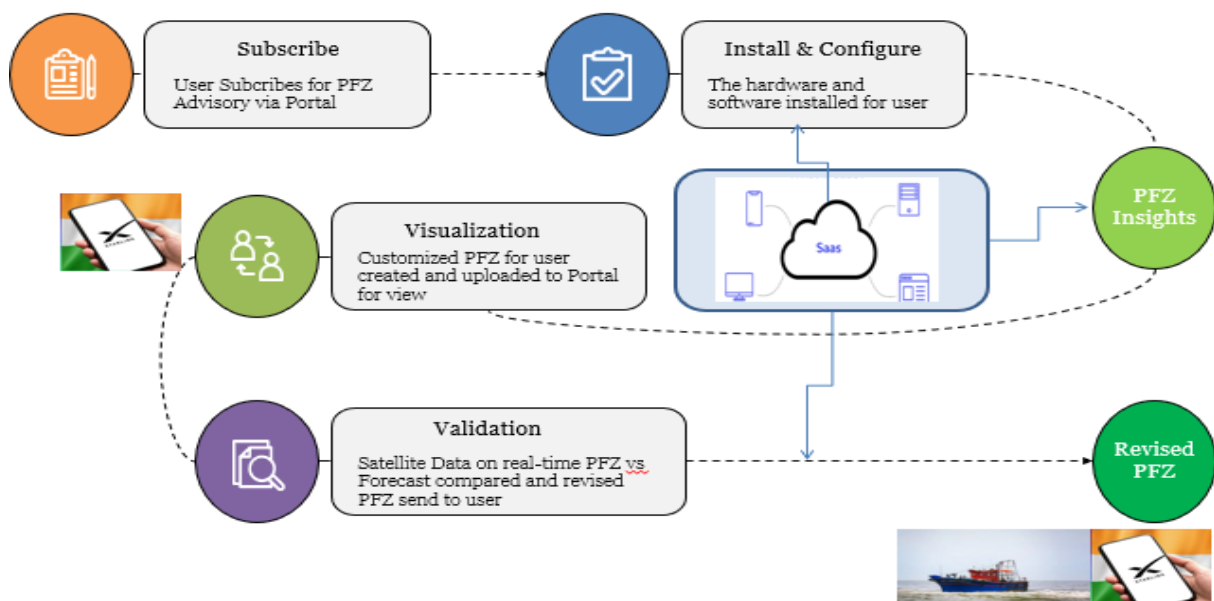
3. Business Context

3.1 Key Business Requirements

- A system that can predict the PFZ in realtime across the coastal area utilizing satellite data
- A system that can forecast PFZ based on historical data and influencing factors in sea conditions
- Network that can enable transmission at sea
- A Hardware device that can display the PFZ information that is transmitted via network
- A validation mechanism that can calibrate the accuracy of forecasting and refine the forecasting

3.1.1 Key Business Processes

The key business process is summarized in the figure below

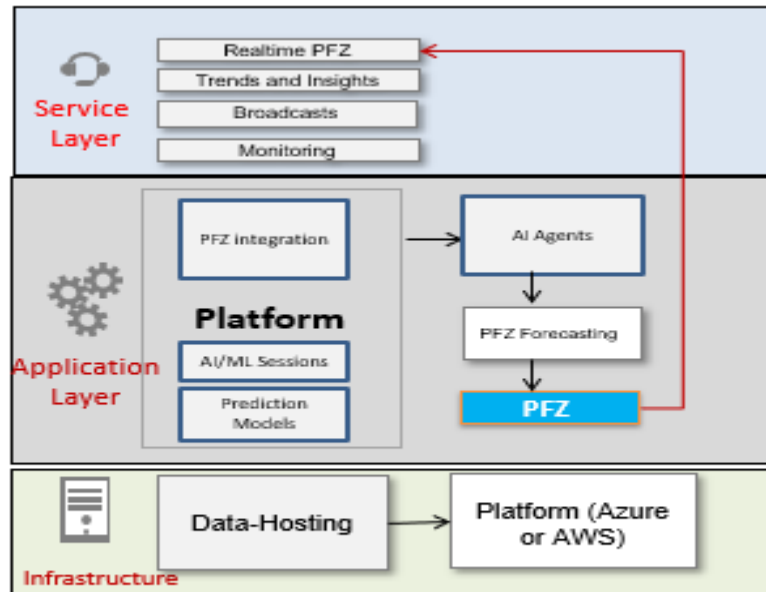


3.2 Non-Functional Requirements

Requirement #	Requirement Description	NFR Category	Implication / Action Taken
NFR.AVAL.001	Scalability: Horizontal scalability	SCALAILITY	
NFR.AVAL.002	High Availability: Auto-healing nodes, multi-region replicas	AVAILAILITY	
NFR.AVAL.003	Latency: Sub-500ms majority	LATENCY	
NFR.AVAL.005	Extensibility: Plugin support	EXTENSIIITY	
NFR.AVAL.006	Disaster Recovery: Multi-region backup, one-click restore	DISASTER RECOVERY	
NFR.AVAL.007	Localization: Multi-language support	EXTENSIBILITY	

4. Conceptual Solution Overview

4.1 Conceptual Architecture



Proposed Platform will have

- Realtime PFZ prediction based on satellite data from INCOIS
- Forecast likely PFZ using historical data
- Forecast Trends based on Historical Data
- Customized PFZ for specific coast

The Platform that is proposed to be built will have

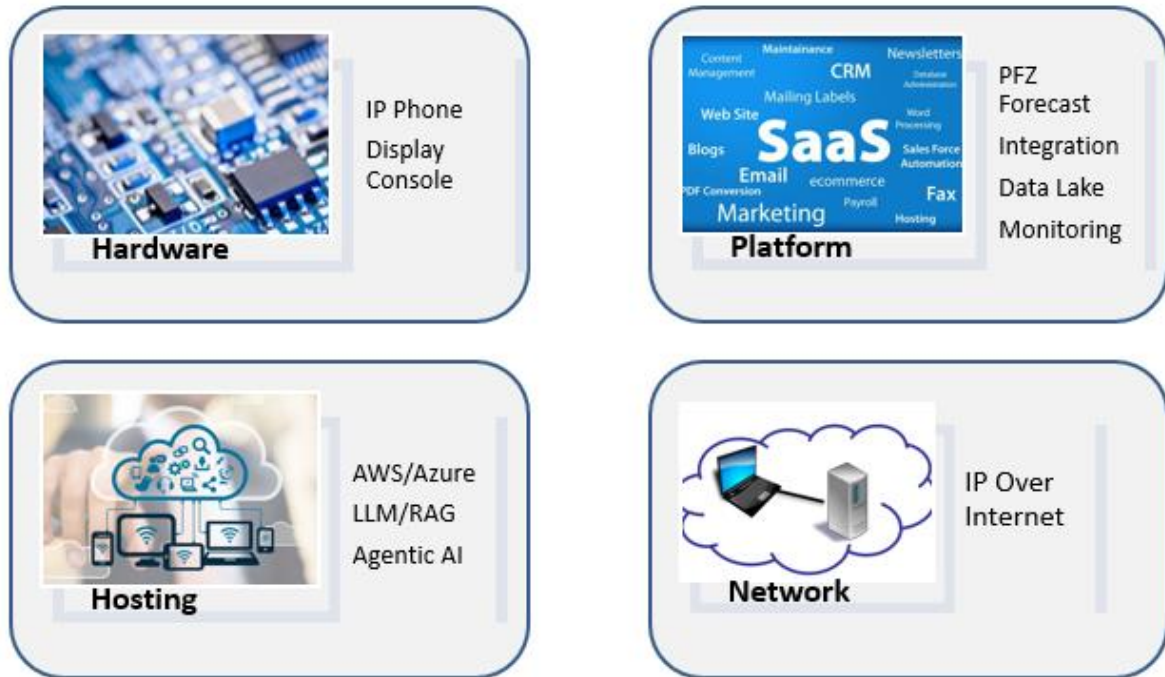
- PFZ forecasting engine
- AI Agents which will be customized for specific coast
- Prediction Models based on LLM and RAG
- Integration with INCOIS and other PFZ data providers

Infrastructure layer will be a robust Cloud hosted solution

- Data-center will be hosted in cloud - Azure or AWS
- The Maintenance of the Cloud Platform will be managed PFZ IT team

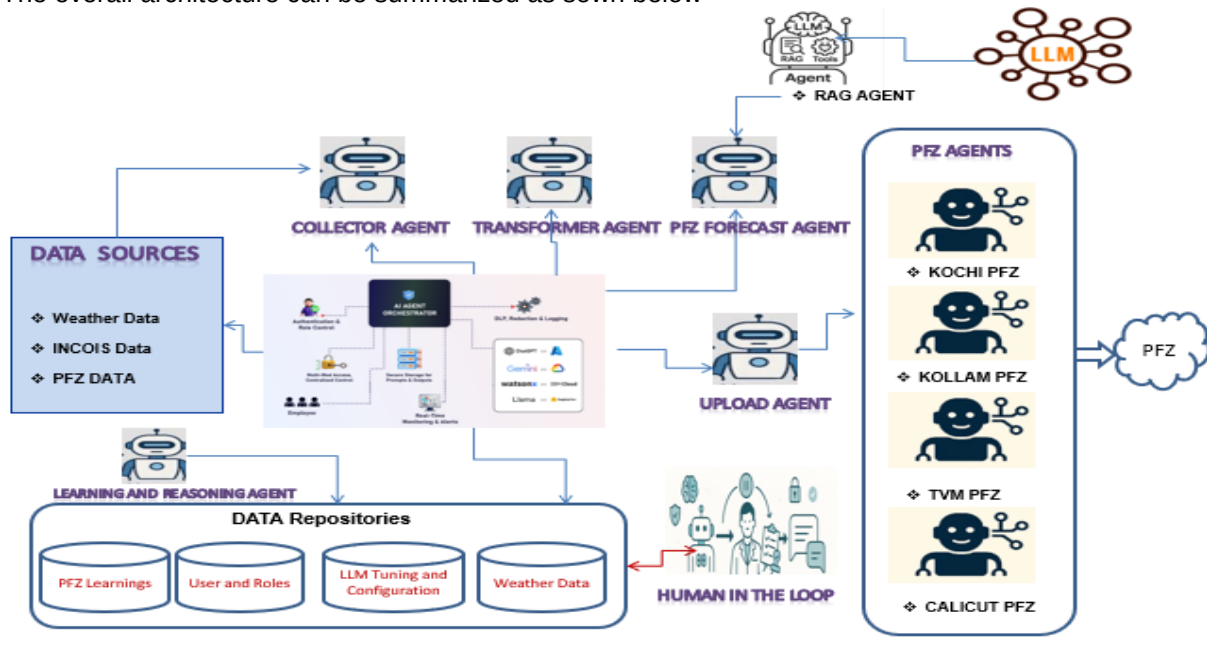
5. Solution Architecture

5.1 Component View



5.2 Application Architecture

The overall architecture can be summarized as shown below



Technology stack consists of

5.2.1 Technology Stack

Layer	Proposed Stack	Remarks
Presentation	○ Mobile App with dashboards ○ PFZ Portal	Available on both Mobile and Laptop
Application	○ PFZ Platform ○ Agentic AI ○ Notification System	
Data	○ PFZ DATA ○ Weather DATA ○ Historical Data	
Integration	○ INCOIS ○ Cloud ○ Social media ○ RAG	
Hosting	AWS Cloud	

5.3 Integration Architecture

5.3.1 Integrating Different Data Sources

PFZ System will be integrated using API-First: OpenAPI for all services, SDK generation, version management

API Design Standards

- OpenAPI 3.1, JSON:API, idempotency, pagination, problem+json errors
- Correlation IDs, traceparent headers, strict versioning
- SDKs: Typescript/C#/Python, Postman collections, Dev Portal examples

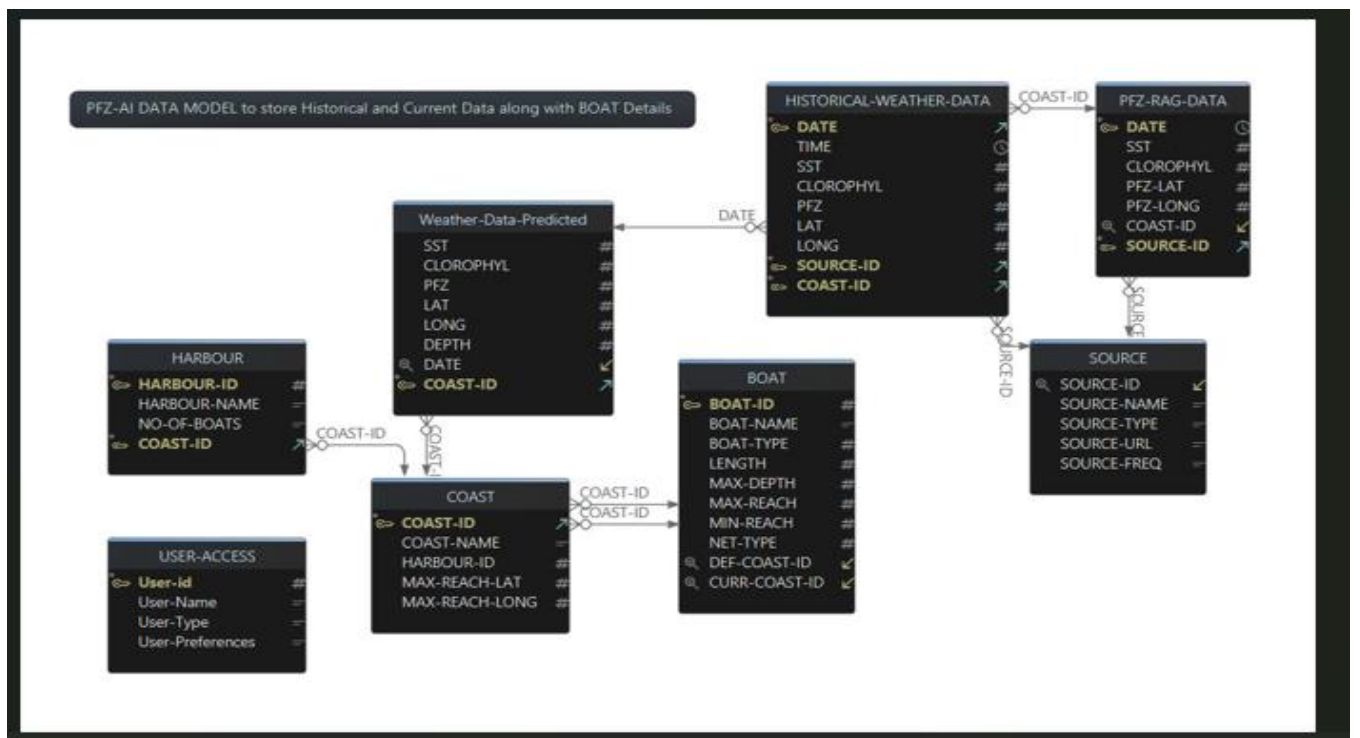
5.3.2 Interface Inventory

Ref.	Interface Name	Desc	Data Object(s)	Source	Target	Freq.	Batch / Realtime
IN1	INCOIS - PFZ	PFZ data	PFZ Data	INCOIS	PFZ Platform	e.g. Daily	e.g. Batch
IN2	INCOIS – Weather Data	Realtime Weather Data	Weather Data	INCOIS	PFZ Platform	e.g. Daily	e.g. Batch

5.3.3 Threat Model

Ref	Threat	Likelihood	Impact	Rating	Action
TRD-01	Validation of approach	High	High	H	Modular rollout
TRD-02	Resource constraints	High	High	H	Finding product development partners , contractors
TRD-03	StartLink/JIO Dependency	High	High	H	Finding right partner

5.4 Data Model



6. Solution Management

6.1 System Operational Management

6.1.1 Backup / Restore / Data Retention Methodology

AWS Cloud based

6.2 User On-boarding

User on-boarding will be through the PFZ-AI Admin screens and includes details related to boat , capacity , nets etc. to enable PFZ platform to provide customized predictions suitable for the boat

7. Solution Delivery Considerations

7.1 Development Considerations

Development tools, language, code repository, versioning, branching, backup, etc.

7.2 Deployment Considerations

- Runtime: App Service, Container Apps, AKS as needed
- CI/CD: GitHub Actions, Azure DevOps, blue-green/canary, DB migrations
- Config: App Config, Key Vault, feature flags

7.3 Data Extraction Considerations

Extracted data from various sources will need to be consolidated in PFZ platform, apply validation and cleansing logic and finally transformed to the target system

8. Appendix

8.1 Open Issues

[list open issues here]